



Radical Behaviorism: A Primer for the Profession and the Person

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Abstract

This article presents radical behaviorism as the philosophical foundation of behavior analysis and argues for its practical relevance to contemporary applied behavior analysis. Written primarily for students and instructors, it is motivated by upcoming credentialing and accreditation standards that will require substantial coursework in behaviorism and philosophical foundations. The article focuses especially on explaining behavior from a radical behaviorist perspective and clarifying the assumptions that underlie the science of behavior analysis. After briefly distinguishing among the conceptual, experimental, applied, and service-delivery aspects of the field, the article traces the historical development of behaviorism from Watson and Thorndike to Skinner's experimental work and his formulation of radical behaviorism. The article emphasizes that radical behaviorism treats behavior—public and private—as physical events subject to lawful relations, rejects mentalistic and dualistic explanations, and interprets verbal behavior and private events (e.g., thinking, consciousness) as environment behavior relations. The core assumptions of radical behaviorism—monism, empiricism, determinism, parsimony, selectionism, and pragmatism—are outlined, and their implications for theory, explanation, and practice are explained. Finally, radical behaviorism is framed as both a professional guide for behavior-analytic activity and a broader worldview that can foster compassion, improve practice, and support cultural and personal change.

Keywords Behavior analysis · Board-certified behavior analyst · Theory · Philosophy · Radical behaviorism

This article examines the philosophy of radical behaviorism as the foundation of behavior analysis, situating it within the broader professionalization of applied

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behavior analysis (ABA). The article is written primarily for students who will be taking courses in radical behaviorism or behaviorism and philosophy because, beginning in 2027, the Behavior Analysis Certification Board (BACB) will require that certificants complete formal coursework, totaling between 45 and 90 contact hr, in Behaviorism and Philosophical Foundations. Such a course will also contribute to the 90 contact hr in Conceptual Analysis, defined broadly as “competence in the history and philosophy of behaviorism, theoretical approaches to understanding behavior, and interpretation of behavior in terms of the concepts and principles of behavior analysis,” that is required for accreditation by the Association for Behavior Analysis International (ABAI). We hope that instructors who teach a course in radical behaviorism will find this article useful to include in their courses as a brief introductory treatment—a primer—of the subject matter. As suggested by the title, our goal in writing this article is not only to provide a relatively accessible introduction to radical behaviorism, but also to show that being a radical behaviorist can enhance your life, both personally and professionally.

According to the Test Content Outline (TCO) of the BACB Task List, Behaviorism and Philosophical Foundations includes five objectives: (1) identifying “the goals of behavior analysis as a science (i.e., description, prediction, control);” (2) explaining “the philosophical assumptions underlying the science of behavior analysis (e.g., selectionism, determinism, empiricism, parsimony, pragmatism);” (3) explaining “behavior from the perspective of radical behaviorism”; (4) distinguishing “among behaviorism, the experimental analysis of behavior, applied behavior analysis, and professional practice guided by the science of behavior analysis”; and (5) identifying and describing the “dimensions of applied behavior analysis.” This article mostly addresses objectives 2 and 3.

However, before moving on, let us briefly address objective number four above, that is, distinguishing “among behaviorism, the experimental analysis of behavior, applied behavior analysis, and professional practice guided by the science of behavior analysis.” Prior to adding service delivery as one of the domains of behavior analysis, there were only three domains: the conceptual (behaviorism or radical behaviorism), the experimental (experimental analysis of behavior), and the applied (applied behavior analysis). The relationship between the (now) four domains of behavior analysis has been discussed (e.g., Hawkins & Anderson, 2002; Moore & Cooper, 2003). Hawkins and Anderson identified four roles that “a behavior analyst might fill: conceptual behavior analyst, basic behavior analyst, applied behavior analyst, and behavior-analytic practitioner” (p. 119).

Moore and Cooper (2003) agreed with Hawkins and Anderson (2002) but conceptualized the distinction between the four domains a little differently. Rather than four domains, Moore and Cooper identified three domains: experimental analysis, applied analysis, and service delivery, all of which they placed on a continuum of behavior-analytic activity mostly with respect to the role of scientific activity carried out by the individual behavior analyst. Important for the present article, Moore and Cooper viewed radical behaviorism as providing “an underlying philosophical position that guides behavior-analytic activity” (p. 80) in the three domains. Thus, for Moore and Cooper, radical behaviorism should not be seen as a separate domain but rather the philosophy that informs the three domains. We agree. Whether you

are a researcher or a practitioner, your practice should be guided by the philosophy of radical behaviorism. The BACB and ABAI apparently also recognize the importance of radical behaviorism as the foundation for the practice of behavior analysis (i.e., service delivery) with their requirements of significant training in behaviorism and philosophical foundations for certification and accreditation, respectively.

A Very Brief History of Behaviorism

John B. Watson is generally credited as being the first psychologist to publicly promote a behavioristic view when he published an article in which he asserted that “psychology as the behaviorist views it is a purely objective experimental branch of natural science” with prediction and control as its theoretical goals (Watson, 1913, p. 158). Watson followed that article with a book titled *Psychology from the Standpoint of a Behaviorist* (Watson, 1919) and then another book some years later, which he revisited and edited, titled *Behaviorism* (Watson, 1930). Unfortunately, Watson’s philosophical call to arms had no real empirical foundation, although by 1916 Watson had come to recognize the relevance of Ivan Pavlov’s research on conditional reflexes. Perhaps most famously, Watson and Rayner (1920) reported that they had established a conditioned emotional response in an infant, Little Albert, by pairing a sudden loud noise with the sight of an otherwise neutral white rat. After the pairing, simply the sight of the white rat reportedly caused Albert to whimper and recoil. Although the experiment by Watson and Rayner has attained mythical status in psychology and has been misinterpreted by many (see Harris, 1979), it does illustrate how Watson latched onto the conditional reflex as the unit of behavior analysis.

By the time Watson began promoting behaviorism, another American psychologist, Edward Thorndike (1911), had already published his *Animal Intelligence: Experimental Studies* in which, among other things, he described his experiments with cats in puzzle boxes, which he had first described 13 years earlier (Thorndike, 1898) and from which he derived his Law of Effect. In addition to Watson and Thorndike, there were rumblings from other experimental psychologists who objected to the subjective approaches that dominated psychology at the time (e.g., introspection) and who were already using objective, experimental methods to study behavior (Malone, 2014; Wozniak, 1997). But it would take another American psychologist, B. F. Skinner, to realize Watson’s call for an objective, experimental science of behavior (e.g., Skinner, 1938).

Early in his career as an experimentalist, Skinner discovered many of the basic principles of operant conditioning that now form the foundation of the science of behavior analysis (e.g., reinforcement, extinction, discrimination, generalization). Over the ensuing decades, he and other experimental behavior analysts built upon that foundation by discovering other principles and refining the existing ones, all through further experimental analysis. But even before Skinner published his first book, *The Behavior of Organisms* (1938), in which he detailed his experimental findings up to that point in time, he was already interested in the philosophy of what was then called behaviorism and was soon to dramatically alter it as a result of his scientific discoveries. Because the main theme of the present article is radical

behaviorism, it is important to understand in more detail what is meant by the term. And because it was B. F. Skinner who coined the term and defined the central tenets of radical behaviorism, we will frequently refer to his writings on the subject.

What Is Radical Behaviorism?

The philosophy of the experimental analysis of behavior is called radical behaviorism. As Skinner (1993) wrote, “I am not sure where the expression ‘radical behaviorism’ came from, but I have always used it to mean the study of behavior without asking about what was going on inside the organism, mental or neurophysiological” (p. 3). But this brief quotation requires clarification. By “inside the organism,” Skinner meant only mental or neurophysiological events because the key feature of radical behaviorism is that events that we call private (stimuli and behavior) are assumed to be physical events that obey the same laws as events that we call public. Of the two—private stimuli and covert behavior—only some private stimuli are literally inside the organism, in the sense organs (e.g., muscles, tendons, joints) and viscera.

Skinner first mentioned radical behaviorism in the commentary section of a series of articles in a special section of the *Psychological Review* (it was called a symposium), which contained Skinner’s contribution, titled “The Operational Analysis of Psychological Terms” (Skinner, 1945). In that article, he made the case that private events (e.g., a toothache) are physical events and could be understood like any other physical event. Moreover, he described how one could learn to talk about (i.e., tact) such private physical events. In one fell swoop, Skinner thus did away with subjectivity in favor of an objective science even of private events. Each author in the symposium was then invited to write a rejoinder, and that is where Skinner first distinguished between what he called “methodological” and “radical” behaviorism (Boring et al., 1945, p. 294). Methodological behaviorists do not consider private events in their scientific analyses because the events are not observed, whereas radical behaviorists do, even, as Skinner wrote, if only inferentially.

That distinction in 1945, however, did not stop others from misrepresenting Skinner’s position as only being interested in observed behavior (e.g., Chomsky, 1959). The difference probably had to do with terminology because Skinner did reject mental (i.e., nonphysical) events. Psychologists who held that there were both physical and nonphysical events were called dualists, whereas Skinner and other radical behaviorists were monists (see below), meaning that for them there were only physical events (see Schlinger, 2018a). Some of what psychologists and others referred to as mental events, Skinner viewed as physical events. For example, much of the time when we say that we are “thinking,” we are simply talking to ourselves (see also Watson, 1920).

At the time, and even today, Skinner’s position on the nature of behavior (*metaphysics*) represented a profound and revolutionary view. As he wrote much later, “A science of behavior must consider the place of private stimuli as physical things, and in doing so it provides an alternative account of so-called mental life. The question then is this: What is inside the skin, and how do we know about it? The answer is, I believe, the heart of radical behaviorism” (Skinner, 1974, p. 233). In this instance,

by “inside the skin,” Skinner just meant unobserved stimuli and behavior. But that was not all Skinner offered in his seminal 1945 article. He also made the first public case that behavior he called verbal was not fundamentally different from other operant behavior; that is, it is shaped and maintained by reinforcement contingencies. That was just a preview of what was to become his major statement on that issue in his book *Verbal Behavior* (1957), which he later called his “most important work” (Skinner, 1977a, p. 379). Thus, a large part of what was to be incorporated into radical behaviorism was this functional approach to language.

The main thrust of Skinner’s (1945) article illustrates how radical behaviorists define terms. We already mentioned that Skinner described how someone’s verbal behavior can come under the control of private events; in other words, we learn to talk about our private events just like we learn to talk about our public behavior. But the more important message in the article is that if one wants to know the meaning of a word, one needs to look at the variables that evoke that word as a verbal response in a speaker. So, for example, if you want to know what a child means when she says “book,” look to see if a book is present as well as an audience (i.e., a listener) who may have reinforced that response in the past or someone who resembles listeners who have reinforced it in the past. If the book and the listener as discriminative stimuli (S^D s) evoke the response “book,” you have discovered the meaning of “book” for that speaker in that one instance. Suppose, however, that the parent says “book,” which then evokes saying “book” by the child (an echoic response). In this instance, the meaning of “book” is different. Thus, from a behavior-analytic perspective, words do not have meanings separate from the circumstances in which they are evoked as verbal responses.

The same analysis can (and should) be applied to scientific terms (Normand, 2019), including behavioral terms such as reinforcement and functional analysis (Schlinger, 2013; Schlinger & Normand, 2013). According to Skinner, what scientists know, what things mean, and what is true are a function of our experiences (Normand, 2019). In particular, as noted above, what a response means is determined proximately by the circumstances under which we say or do something (i.e., establishing operations [EOs], S^D s) and ultimately by our learning history (i.e., reinforcement) that gives rise to the control by those proximate events (see Schlinger & Blakely, 1994). This also applies to the person behaving and to those observing that person, including any scientists who might be studying the behavior.

But that was not all Skinner included in his article. At the end, in one profound paragraph, Skinner offered one solution to the problem of consciousness or self-awareness when he wrote, “Being conscious, as a form of reacting to one’s own behavior” (verbally), “is a social product” taught to us by others, and “[t]he individual becomes aware of what he is doing only after society has reinforced verbal responses with respect to his behavior as the source of discriminative stimuli” (Skinner, 1945, p. 277). This was, and still is, another revolutionary or “radical” idea and an example of a radical behavioral theory of a traditional topic—consciousness or self-awareness—that psychologists and philosophers have constantly struggled with. More recently, other behavior analysts have elaborated upon and extended a radical behavioral view of what we mean by *consciousness* or *self-awareness* (e.g., Leslie, 2015; Schlinger, 2008, 2009a). But even with terms such as consciousness (or

memory, perception, emotion), rather than debating their meanings or definitions, the radical behavioral position is to look at the variables (EOs, S^Ds) that cause one to say that someone is, for instance, “conscious.” Skinner’s (1945) article was groundbreaking in many ways, and it established many of the tenets of what we now know as radical behaviorism.

Since Skinner introduced the philosophy of the science of behavior analysis called radical behaviorism, numerous authors have written articles and books on radical behaviorism. For example, there are books on behaviorism by major figures in the field, such as Baum (2017), Chiesa (1994), Moore (2008), Rachlin (1991), Smith (1986), Staddon (1993), and Zuriff (1985). Each of these authors has his or her own take on (radical) behaviorism, each reflecting his or her own training and experience, but they all share some common assumptions and beliefs.

The “Radical” in Radical Behaviorism

The term *radical* in radical behaviorism warrants some attention in and of itself. Following from the discussion above about the meanings of terms, in general we should take the same approach to behavioral terms (Schlinger, 2013). Thus, we find that the term *radical* has been used somewhat differently by various authors (see Schneider & Morris, 1987, for a detailed discussion of the history of the term *radical behaviorism*). There seem to be at least four definitions or uses. First, and most central, is that *radical* “implies a thoroughgoing and comprehensive behaviorism, rather than an extreme position that is fanatically restricted to only publicly observable behavior” (Moore, 2011, p. 456; see also Cooper et al., 2020). To wit, *thoroughgoing* and *comprehensive* mean that radical behaviorism includes all behavior, observed and unobserved, as well as behavior Skinner called verbal. This use seems to be the most consistent with Skinner’s writings.

A second, and related, use of *radical* is “of or relating to the origin; fundamental” (Merriam-Webster, *Collegiate Dictionary*, 11th ed.). In this sense, *radical* refers to tracing both observed and unobserved behaviors to their origins and explaining them in terms of fundamental processes of natural selection and conditioning (Malagodi, 1986). Third, *radical* is an appropriate adjective because radical behaviorism represents a dramatic break from other ways of thinking about behavior, most notably those that posit mental structures and processes. Finally, Malagodi (1986) proposed that in addition to explaining public and private events, radical behaviorism, as a world view, “attempts to interpret all dimensions of human behavior, including those traditionally assigned to the fields of economics, political science, education, social philosophy, etc.” because these enterprises “constitute the sources of culturally important human behaviors” (p. 4). Thus, from Malagodi, comes a fourth use of the term *radical*, one that favors “drastic political, economic, or social reforms” (p. 15).

Skinner’s (1989) later definition of radical behaviorism as “the philosophy of a science of behavior treated as a subject matter in its own right apart from internal explanations, mental or physiological” (p. 122) is slightly different from the way he described radical behaviorism 44 years earlier in his 1945 article. In 1945, radical

behaviorism was about not ruling out internal events in the analysis of behavior. The only caveat was that such events were physical, not mental. By 1989, Skinner had had time to ponder a broader meaning of radical behaviorism, as “the philosophy of science underlying behavior analysis, the science of behavior, and its application” (Moore, 2008, p. 4). This broader view would seem to include Malagodi’s (1986) proscription to include addressing culturally and socially important human endeavors.

What Is Behavior?

Behavior, broadly defined, and its environmental causes, constitute the subject matter of behavior analysis. As noted previously, and as Skinner (1963) has pointed out, “The skin is not that important as a boundary. Private events have the same kinds of physical dimensions” (p. 953). He meant energy changes (i.e., stimuli) are the same whether they originate inside the body (e.g., a toothache) or outside the body (e.g., the sound of a dog barking). Also, some behaviors can be easily observed and measured, and other behaviors cannot at present be observed and measured. It would not be correct to call the latter “inner” behavior or say that it is occurring “in the mind.” It is simply occurring at a level that cannot be directly observed and measured or is below what Palmer (2009) called “the threshold of observability,” which he defined as “that set of conditions under which the response is just detectable” (p. 12). Consider that most of the time when we say we are “thinking,” what we mean is that we are talking to ourselves. Sometimes when we talk out loud to ourselves—for example, when we are alone or when we have a particularly tricky problem to solve—and someone hears us, we report that we were just “thinking out loud.” The behavior of self-talk is usually called covert, private, or, more accurately, subaudible, even though as described above, it can easily become audible under certain conditions. Likewise, radical behaviorists have noted that when we imagine hearing someone’s voice or a piece of music, we are really talking or singing to ourselves subaudibly. Moreover, there is neuroimaging evidence that supports such a theory. When people report imagining hearing speech or music, the language and motor areas of the cortex become active, supporting the notion that they are talking or singing to and “hearing” themselves (Schlinger, 2009b).

Behavior as a Subject Matter in Its Own Right

In a chapter titled “The Behavior of Organisms at Fifty,” Skinner (1989) wrote, “I don’t believe I coined the term radical behaviorism, but when asked what I mean by it, I have always said, ‘the philosophy of a science of behavior treated as a subject matter in its own right apart from internal explanations, mental or physiological’” (p. 122). But what did Skinner mean by “behavior treated as a subject matter in its own right?”

Radical behaviorism is to be distinguished from cognitive psychology, which is mostly interested in behavior for what it means about some mental, cognitive, or, nowadays, even physiological process. For example, psychologists have conducted

countless experiments on memory, but not once have they ever directly observed the memory. What they do directly observe and measure is behavior (e.g., recalling lists of numbers, letters, or words) and sometimes correlated physiological events. But cognitive researchers are not as interested in the actual recall behaviors and what might cause them, or even in the correlated physiological events underlying such behavior, as what the behaviors allow them to say about constructs such as short- and long-term memory, for example. Behavior analysts, in contrast, are interested in the behaviors that are observed and measured and the variables that cause them to occur—those that we call *memory* or *remembering* (see Palmer, 1991).

Contrary to some critiques of behaviorism suggesting that behaviorists dismiss as unimportant biological or physiological events as unimportant (e.g., de Waal, 1999; Pinker, 2002), behaviorists fully acknowledge the importance of such events. In fact, Skinner acknowledged the role of physiological events as early as 1938 (see also Morris et al., 2004). Behavior analysts view such neural events as proximate (or proximal), not ultimate causes of behavior (Alessi, 1992; Schlinger, 2015), where proximate refers to causes close in time to behavior and include variables such as S^Ds, EOs, and CSs, and where ultimate causes are further back in time (i.e., historical) and include the evolutionary history of the species and the individual's unique learning history to explain why the proximate causes function as they do. But behavior analysts mostly leave the study of neural events and their relationship to behavior to neuroscientists. As Skinner (1953) noted, any new discoveries in neuroscience would not invalidate the laws of behavior, but he also acknowledged that a complete picture of behavior would ultimately require an understanding of the relevant physiology. To conclude, as Schnaitter (1999) wrote,

Behavior is not simply data; nor is it a symptom of an underlying mental state, nor is it a mere index of events taking place at some other place or in some other dimensional system. Behavior is genuinely the subject in its own right. A given instance of behavior is taken to be explained when its occurrence is shown to stand in a functional relationship to one or more aspects of the environmental context in which the behavior has been occurring. The primary causal mode of the analysis is selection by consequences. (p. 213)

Assumptions of Radical Behaviorism

Radical behaviorism rests on several philosophical assumptions about the nature of behavior and how it should be studied scientifically.¹ These assumptions guide both the experimental analysis of behavior in laboratory settings and the application of behavioral principles to practice areas such as education, therapy, and organizational behavior management. The major assumptions commonly discussed in behavior

¹ For more on the assumptions of radical behaviorism, see Delprato and Midgley (1992), as well as Moore (2008), who referred to these as “basic principles of Skinner’s radical behaviorism” and Cooper et al. (2020), who called some of them (determinism, empiricism, and parsimony) “attitudes of science.”

analysis are monism, empiricism, determinism, parsimony, selectionism, and pragmatism (e.g., Baum, 2017; Cooper et al., 2020; Johnston, 2021; Moore, 2008, 2011; Skinner, 1953, 1969, 1974). Although these assumptions are not themselves directly testable in the way that specific hypotheses are tested experimentally, they function as guiding principles for the science of behavior. In other words, they are epistemological, not ontological. They shape how behavior analysts interpret data, design experiments, develop interventions intended to improve socially significant behavior, and, indeed, view the world of behavior at large. Skinner (1974) argued that a coherent science of behavior requires such philosophical commitments because they determine what counts as an acceptable explanation and how scientific inquiry proceeds. In other words, they are necessary for a pragmatic science (see below).

Monism

Monism (also called *naturalism*; see Delprato & Midgley, 1992) is the assumption that behavior and the events that influence it are part of the physical world. For example, the statement, “Cognitive psychology is concerned with explaining the structure and mental operations of intelligence as well as its behavioral manifestations” (Frensch, 2001, p. 2147) is dualistic, as is the question, “How does the nonmaterial mind influence the brain, and vice versa?” (Fischbach, 1992, p. 48). Statements and questions such as these divide the world into two parts: the mental (or cognitive) and the physical (or behavioral). Skinner emphasized that explanations of behavior should focus on relations between behavior and the environment rather than on hypothetical mental entities, thus clarifying that mental events are not private events.

Radical behaviorism rejects mind–body dualism (sometimes called Cartesian dualism, after the philosopher René Descartes)—the view that mental events belong to a separate, nonphysical realm—and instead treats both public and private events as physical processes. As Skinner wrote, “Private and public events have the same kinds of physical dimensions” (Skinner, 1963, p. 953). Thoughts, feelings, and other private events are therefore not causes of behavior but are themselves viewed as behavioral or physiological events subject to environmental histories (Skinner, 1953, 1974). For example, as we mentioned previously, when we say we are thinking, we are usually talking to ourselves. The self-talk is mostly unobserved by others. But sometimes we think (e.g., talk to ourselves) out loud. Thus, the distinction between public and private events reflects differences in accessibility (i.e., observability) rather than differences in kind; both belong to the same natural world and, therefore, fall within the scope of scientific analysis.

Radical behaviorism does not deny private events but interprets them as stimuli and behavior that are only different than public stimuli and behavior in that they are presently unobserved. Behavior analysts in practice inherently assume monism when they talk about a client’s private events (e.g., feeling and thinking) as just more stimuli and behavior, with the only exception being that they are unobserved and, thus, inaccessible. However, sometimes ABA researchers can get verbal clients

to talk aloud as they solve problems, which moves the behavior from the covert to the overt (see Austin & Delaney, 1998).

Empiricism

Empiricism is the assumption that knowledge about behavior must come from systematic observation and experience. From a radical behavioral perspective, *experience* involves “the acquisition, maintenance, and modification of the behavior of organisms through their interaction with the environment, as well as the subsequent alteration of that environment through such interaction” (Marr, 2003, p. 64). In behavior analysis, this principle means that relations between behavior and environmental variables are discovered through careful measurement, experimentation, and replication. Skinner (1953) described behavior as something that changes through interaction with the environment: organisms act upon the world, and the consequences of their actions alter the probability that those actions will occur again.

Because of its empirical foundation, behavior analysis emphasizes objective and reliable measurement. Through repeated observation and experimental control, scientists develop general principles that describe how behavior changes under different conditions (Skinner, 1953, 1969). Empiricism is “the foremost rule in behavior analysis” as it relates to “the practice of objective observation and measurement” of environment-behavior relations (Cooper et al., 2020, p. 5). In research and practice, behavior analysts collect observable data on behavior and environmental conditions to identify functional relations between them. When behavior analysts in practice observe behavior, collect data, and then design interventions based on those observations, they are taking an empirical approach to the problem. Contrast that with practitioners who simply do what they think is best or what they have been trained to do without actually observing and measuring behavior.

Determinism

Determinism is the assumption that behavior is lawful and orderly and that events occur as the result of potentially identifiable causes. Behavior analysts assume that behavior is influenced by genetic endowment, environmental history, and current environmental conditions. This assumption allows scientists to search for the variables responsible for behavior and to discover predictable patterns in how behavior changes. Skinner (1974) argued that assuming determinism is essential for scientific inquiry. As he wrote, “Science is more than the mere description of events as they occur. It is an attempt to discover order, to show that certain events stand in lawful relations to other events.... If we are to use the methods of science in the field of human affairs, we must assume that behavior is lawful and determined” (Skinner, 1953, p. 6). If behavior were fundamentally random or uncaused, attempts to analyze and control it would be futile. Instead, the discovery of functional (causal) relations between behavior and its environment allows researchers to control and predict behavior effectively.

The assumption of determinism in the analysis of behavior is reflected in a common expression: “The rat is always right.” The same point can also be made with reference to the pigeon or student always being right (e.g., Bachman & Farris, 1968; Findley, 1962; Pliskoff & Gollub, 1974; Skinner, 1977b).² Although the phrase might seem amusing, it has important philosophical implications for how behavior is viewed within radical behaviorism. Consider an anecdote common from laboratory courses: students each shape a rat’s lever pressing and bring the response under discriminative control of a light before arranging various schedules of reinforcement. It is inevitable that one student claims to have received the “dumb” rat because, whereas the other rats learn to press the lever, their rat lies motionless in the corner of the chamber. But closer inspection reveals that the student had been delivering food pellets when the rat was in that corner to draw it toward the food cup near the lever, inadvertently shaping the problematic corner-dwelling. From a behavioral perspective, the rat was behaving exactly as its reinforcement contingencies shaped it to behave—the rat was right—whereas the student’s actions were likewise determined by their own reinforcement history and, thus, the student was right too (see also Keller, 1968).

In practice, and in our everyday lives, radical behaviorists assume that all behavior has causes and that, for the most part, those causes can be found in the individual’s reinforcement history and current environment as defined above. We may not like the behavior we observe, but we understand that it is caused and not the result of the individual’s choice or intention or any other invented mentalistic construct. It is ironic that this deterministic approach to behavior leads to a more compassionate view of individuals because we do not blame or hold them responsible for their behavior.

Parsimony

The principle of parsimony in science has a long history (for a brief description, see Schlinger, 2018b). Parsimony refers to the preference for explanations that rely on the fewest assumptions while still accounting for the observed data. In behavior analysis, this means favoring explanations based on directly observable environmental contingencies before appealing to more complex or hypothetical constructs. The goal is not to oversimplify behavior, but to avoid unnecessary speculation. Such speculation invoking unseen mental processes can obscure the environmental variables that are actually responsible for behavior. By beginning with the simplest adequate explanation, scientists increase the likelihood of identifying variables that can be manipulated experimentally and used to change behavior effectively. In practice, a parsimonious approach suggests that we look to environmental variables that can be observed and tested to explain an individual’s behavior before speculating about unobserved mental or emotional processes (e.g., anxiety) or more complex, unseen circumstances operating outside of the situation.

² We thank Ed Morris for these references.

By focusing first on measurable relations between behavior and environment, scientists are more likely to identify variables that can be manipulated and tested. Behavior analysts, therefore, begin with the simplest explanation consistent with the evidence, focusing on the relationship between behavior and its consequences. A classic example comes from laboratory research. When a rat presses a lever in an operant conditioning chamber and receives a food pellet, the increased rate of lever pressing is explained by the reinforcing effect of the food consequence rather than by assuming that the rat presses the lever because it “expects” or “wants” food or has developed a mental understanding of the task. Such explanations, however, point to processes or structures that cannot be independently observed, measured, or tested (e.g., Schlinger, 1993), that is, they are not empirical. These kinds of explanations of behavior are also circular, or what Skinner (1957) called explanatory fictions. They are circular because the only evidence for the explanation is the behavior to be explained the only evidence of the rat expecting the food is that it presses the lever.

By identifying the environmental contingencies responsible for the behavior, researchers can predict and reliably reproduce the effect under similar conditions (Skinner, 1969). In practice, when we explain that a child’s tantrums are maintained by contingent parental attention when the child throws a tantrum, as opposed to appealing to invented constructs (e.g., anger or frustration), we are offering a parsimonious explanation of the behavior; that is, one that points to observable, measurable, and testable events.

Selectionism

In his book *Science and Human Behavior* (1953), Skinner formally introduced the idea that operant learning represents a kind of selection process. Years later, he published the article “Selection by Consequences” (Skinner, 1981) in which he again described similarities between natural selection and behavioral selection. Selectionism is the view that behavior develops through processes of variation and selection by consequences, and Skinner drew a parallel between biological evolution and operant conditioning: just as genetic variations are selected through survival and reproduction, behavioral variations are selected through reinforcement. This perspective recognizes multiple levels of selection. Natural selection shapes species characteristics across generations, operant selection shapes behavior within the lifetime of the individual through reinforcement, and cultural selection shapes practices maintained within groups because they produce beneficial consequences.

In evolutionary terms, the concept of selection was revolutionary because it countered the prevailing belief that traits were fixed, which meant that there was a need to speculate about who or what created those traits. In behavioral terms, the concept of selection by consequences was also revolutionary in that it countered the still-prevailing view that behavior comes from within the individual or occurs freely. When behavior technicians or therapists are sitting in front of a client, a potentially infinite number of responses are available from which to choose (i.e., select) with a reinforcing consequence. The therapist can prompt or instruct a response or can extinguish

an already reinforced behavior to generate some varied responses from which to choose (i.e., select) by providing a reinforcing consequence. Since Skinner's (1981) article, the selection of behavior has become a core assumption of radical behaviorism (see Donahoe, 2003; Donahoe & Palmer, 1994).

In practice, anytime a behavior analyst in practice reinforces a response, they are selecting similar responses to the exclusion of other, alternative, responses. This procedure is known as differential reinforcement, and it is used in the procedure of shaping, in which closer—successive—approximations to some target behavior are selected. Of course, differential reinforcement and shaping are not limited to intentional programs to change behavior; our behavior is selected every day by consequences as we interact with our natural environment.

Pragmatism

Pragmatism emphasizes the practical value of scientific explanations. In radical behaviorism, explanations of behavior are evaluated partly by their usefulness in controlling and predicting behavior. Knowledge is meaningful when it contributes to effective action (Moore, 2011; see also Lattal & Laipple, 2003). Skinner (e.g., 1953, 1974) emphasized that the goal of a science of behavior is not merely to describe behavior, but also to develop technologies that improve human life. Applications in education, therapy, and social systems demonstrate the practical value of behavioral principles.

One of Skinner's most enlightening and entertaining illustrations of his pragmatic approach to science is his article "A Case History in Scientific Method" (1956) in which he described how he came to discover reinforcement, extinction, and schedules of reinforcement. He was simply looking for order in the behavior of an individual organism (determinism). In the process of doing that, he derived five informal principles of scientific method, including, (1) when you run into something interesting, drop everything and study it; (2) some ways of doing research are easier than others; (3) some people are lucky; (4) sometimes apparatuses break down; and (5) serendipity, in which a researcher finds one thing while looking for another. All these informal principles were derived from the pragmatic outcomes of Skinner's activities; that is, they resulted in Skinner being able to act more practically in the sense that he could more effectively control and predict behavior, that is, discover its orderliness. The concepts and principles of behavior analysis are pragmatic if they can lead to effective action—control and prediction—both in experimental and applied settings.

Radical Behaviorism and Theory

Radical "behaviorism is the philosophy of science underlying the science of behavior. It takes behavior as a subject matter in its own right and applies the same principles and methods of other natural sciences to develop theories and

explanations” (Moore, 2011, p. 460). What, then, is the relationship between theories and explanations in behavior analysis? In short, theory in behavior analysis is like theory in the other natural sciences (e.g., biology, geology, chemistry, physics). In other words, it is a set of statements (or equations) that have been induced by the discovery of laws (or principles) that are then used to deduce observations not amenable to experimental analysis. Although the natural sciences of physics, chemistry, and biology are now all theoretically rich enterprises, they began more simply with scientists trying to answer nontheoretical questions about the discovery of what are called functional relations between independent and dependent variables. More simply, these scientists were trying to figure out how the world works.

In everyday terms, the term *theory* is used to mean guess or hypothesis, as in “What is your theory of why that child throws tantrums?” This is another way of saying “What is your guess about why that child is throwing a tantrum?” But theory in science is not an educated guess. According to Skinner (1947),

Theories . . . are statements about organizations of facts. The atomic theory, the kinetic theory of gasses, the theory of evolution, and the theory of the gene are examples of reputable and useful scientific theories. They are all statements about facts, and with proper operational care they need be nothing more than that. But they have a generality which transcends particular facts and gives them a wider usefulness. Every science eventually reaches the stage of theory in this sense. (p. 28)

Thus, theory in behavior analysis consists of explanations based on the basic principles of operant and respondent conditioning about novel behavior, or about behavior that has not or could not be subjected to laboratory analysis. For example, suppose you see a child throwing a tantrum in a grocery store after the child asks for candy and the parent says no. You observe that the parent gives the child a candy and then the child calms down. Your friend asks you why you think the child throws the tantrum and, perhaps, why the parent would give the child candy for doing so. Based on your knowledge of reinforcement and, perhaps, stimulus control, you suggest that getting the candy immediately after throwing a tantrum probably reinforces the tantrum behaviors and that the cessation of the tantrum after getting the candy probably reinforces the parent’s behavior of giving the candy under those circumstances. What you mean, of course, is that the next time that parent and child are in that or a similar (generalization) situation, you predict that both the child and the parent will behave in a similar manner. Now, of course, it would be difficult to take such behavior into the laboratory, although you could do a functional analysis in a treatment setting. But as you are a casual observer of this particular interaction, you are essentially using behavior analytic theory, which appeals to the basic principles to explain the child’s behavior of throwing a tantrum and the parent giving candy. If you view all behavior through this theoretical lens, then you are acting as a radical behavior analyst.

Skinner’s book *The Behavior of Organisms: An Experimental Analysis* (1938) and his article “The Operational Analysis of Psychological Terms” (1945) ushered in an operant theory of behavior and radical behaviorism as the philosophy of the

experimental analysis of behavior. By the time he published his 1969 book *Contingencies of Reinforcement: A Theoretical Analysis*, Skinner was steeped in behavior-analytic theory. And contrary to the claim by some that Skinner was anti-theory, in the article which was the impetus of such claims, “Are Theories of Learning Necessary?” (1950), Skinner acknowledged the fruitfulness of theory in science and noted that all statements, even scientific statements, are theoretical in the sense that they are never certain, just varying in degrees of confidence.

Radical Behaviorism as a World View

There are at least two ways in which talking about radical behaviorism as a world view can be understood, both of which essentially involve how individual radical behaviorists talk about behavior. The first was suggested by Malagodi (1986). To repeat, he listed three definitions of *radical*: (1) of or pertaining to roots or origins; fundamental; (2) thoroughgoing or extreme; and (3) favoring drastic political, economic, or social reforms” (p. 15). He then called for radical behaviorism to be conceptualized as a “thoroughgoing world view” that encourages social reform by incorporating some principles from cultural evolutionary theory (e.g., Glenn, 1985; Harris, 1980). Many radical behaviorists, beginning with Skinner (e.g., 1948, 1953, 1961, 1971, 1978) have encouraged social reform through radical behaviorism (e.g., Biglan, 2015; Glenn, 2004; Mattaini et al., 1996). The point is that, for many, social reform is or should be a tenet of radical behaviorism.

The second way in which we may talk about radical behaviorism as a world view is more personal. As this article’s authors, we hold radical behaviorism as a world view. But what does that mean? In short, it means that we not only view practicing behavior analysis through a radical behavioral lens, but we also view behavior in general through that lens. We (try to) approach the behavior of our children, spouses, parents, friends, and pets from the viewpoint of radical behaviorism. We also view the behavior of politicians, clergy, and celebrities from such a viewpoint. We hold to the philosophy that individuals are always right, which does not mean that we always like or approve of what they do, but rather that their behavior is determined by their reinforcement history and current circumstances, as well as their genetic endowment. Because of our respective training histories, and years of practicing and talking like radical behaviorists, viewing the world of behavior this way is second nature to us.

A radical behavioral *Weltanschauung* can be liberating in the sense that one has a consistent way of viewing behavior without the trappings of mentalism and free will. But it can also be a cross to bear because one is now in the position to understand, at least in principle, why people behave as they do. On the one hand, much of the time we cannot do much about the behavior of others because changing another person’s behavior requires control over at least some of the variables responsible for it. Viewing the world of behavior as a radical behaviorist means that there is no more blissful ignorance. On the other hand, even when cynicism is the easiest reaction to the problems we face, a radical behavioral world view still offers some optimism that behavior can change and that a radical behavioral view can show us where to look to make the

change—behavior and the environment. For those who truly understand radical behaviorism, the benefits are life-changing and lifelong.

Conclusion

In conclusion, radical behaviorism remains the philosophical foundation of behavior analysis and an essential framework for both the profession and the person. As the field continues to mature and formal training in behaviorism and philosophical foundations becomes a requirement for certification and accreditation, students and practitioners alike will benefit from understanding the assumptions that underlie behavior science. Radical behaviorism is not merely a historical doctrine attached to Skinner's name; it is a coherent philosophy of science that treats behavior as a subject matter in its own right, includes both public and private events within a naturalistic framework, and offers a disciplined alternative to mentalistic explanations. Its assumptions—monism, empiricism, determinism, parsimony, selectionism, and pragmatism—guide scientific inquiry and practical intervention alike. At the same time, radical behaviorism is more than a professional orientation. As a world view, it encourages a compassionate understanding of human action that avoids blame and directs attention toward the environmental and historical variables that make behavior lawful and changeable. For students entering the field, this primer is intended as an invitation to deeper study. For practitioners, it is a reminder that effective practice is strongest when grounded in a clear and consistent philosophy of behavior.

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